

A prospective study of diet and adult-onset asthma.

A role for diet in the pathophysiology of asthma may be mediated by altered immune or antioxidant activity with consequent effects on airway inflammation. We evaluated associations between several dietary factors assessed by a semiquantitative food frequency questionnaire, and incidence of asthma over a 10-yr period in 77,866 women 34 to 68 yr of age. Women in the highest quintile of vitamin E intake from diet, but not from supplements, had a risk of 0.53 (95% confidence interval [CI] = 0.33 to 0.86) compared with women in the lowest quintile. This relationship, however, was attenuated when the contribution from nuts, a major source of vitamin E in these data and a possible allergen, was removed (relative risk = 0.74 [0.50 to 1.10], p for trend = 0.007). Positive associations were found for vitamins C and E from supplements, but appeared to be explained by women at high risk of asthma initiating use of vitamin supplements prior to diagnosis. A nonsignificant inverse association with carotene intake was noted, but no clear relations with asthma were demonstrated for intake of linoleic acid or omega-3 fatty acids. These data suggest that antioxidant supplementation and intake of various fats during adulthood are not important determinants of asthma, although vitamin E from diet may have a modest protective effect.